CDT204 – Computer Architecture

> Date: Aug. 21st 2025
>
> Time: 8:30 – 12:30
>
> Help: Small calculator is allowed in the exam

The exam has 71 points. The grades will be awarded as follows:

- 3 : 36 Points
- 4 : 49 Points
- 5 : 63 Points

Important Notes:

- Give as full an answer as possible to obtain full marks. All calculations, approximations, assumptions and justifications must be reported for full credit unless stated otherwise. Please use figures and examples to clarify.
- If you do not understand a question clearly, you are allowed to call the teacher and ask.
- Write the question and part number on each page clearly.
- Answer each question on a separate sheet.
- In case you might have forgotten:

  1GB = 2^{10}MB = 2^{20}KB = 2^{30}B

  1 sec = 10^3 ms = 10^6 μs = 10^9 ns = 10^{12} ps

  $$S = \frac{T_0}{\frac{T_A}{F} + (T_0 - T_A)}$$

Live long and prosper

Task 1 (10p)

a) A program spends 80% of its time doing multiply instructions. If we can redesign the multiplier in the pipeline in a way that it gets 4x faster, how much faster does the program run on this new design? (**2p**)
b) What is the limit on the maximum speedup if the multiplier from part (a) is made infinitely fast? (**2p**)
c) For a 4-bit two's complement system, show the following operations in binary: (**6p**: a(3p), b(3p))
    a. 4-7
    b. 8-6

Task 2 (14p)

When a program is adapted to run on multiple processors in a multiprocessor system, the execution time on each processor is comprised of computing time and the overhead time required for locked critical sections and/or to send data from one processor to another.

Assume a program requires $t = 500s$ of execution time on one processor. When run on p processors, each processor requires t/p seconds, as well as an additional $4s$ of overhead, irrespective of the number of processors.

a) Compute the per-processor execution time for 2, 4, 8, 16, 32, 64, and 128 processors. For each case, list the corresponding speedup relative to a single processor and the ratio between actual speedup versus ideal speedup (speedup if there was no overhead). (**7p**)
b) Given infinite number of processors. What would be the maximum theoretical speedup that we can gain? (**2p**)
c) Draw a chart with the number of processors on the x-axis and the speedup on the y-axis. Plot both actual speedup and ideal speedups that you calculated for the previous part into this graph. Discuss your findings. (**5p**)

Task 3 (15p)

Consider a hypothetical machine called SIC, for Single Instruction Computer. As its name implies, SIC has only one instruction: *subtract and branch if negative*, or **sbn** for short. The **sbn** instruction has three operands, each consisting of the address of a word in memory:

```
sbn a, b, c     #Mem[a] = Mem[a] - Mem[b]; if (Mem[a]<0) go to c
```

The instruction will subtract the number in memory location b from the number in location a, and place the result back in a, overwriting the previous value. If the result is greater than or equal to 0, the computer will take its instruction just after the current instruction. If the result is less than 0, the next instruction is taken from memory location c. SIC has no registers and no instructions other than sbn.

A. Identify what the following code does and write the equivalent code in C (The ".+1" in the code means "The address after this one") (**4p**).

```
start: sbn temp, temp, .+1
       sbn temp, a, .+1
       sbn b, b, .+1
       sbn b, temp, .+1
```

B. Assume that Mem[0] has a constant value of zero and Mem[1] has a constant value of one (they both cannot be written to). Also assume that memory locations 2 to 8 can be used for storing temporary values (main program and its data will not be stored there). How could the following MIPS instructions be done in SIC code? (**11p**: a(3p), b(4p), c(4p))
   a. J
   b. ADD
   c. MOVE

Task 4 (8p)

Consider two different implementations M1 and M2 of the same instruction set. There are four classes of instructions (A, B, C, and D) in the instruction set. M1 has a clock rate of 3.5 GHz and M2 has a clock rate of 4.5 GHz. For a given program, the fraction of all instructions that belong to a particular class, and the average number of cycles for each instruction in the two implementations are as below:

Class	Percentage of instructions	CPI on M1	CPI on M2
A	10%	1	2
B	35%	2	2
C	25%	2	4
D	30%	3	4

A. Which implementation on the instruction set can run the program faster? By how much? (**5p**)
B. What should be the clock speed of the slower implementation so that it can have the same speed as the faster implementation? (**3p**)

Task 5 (12p)

Assume a cache design with 32-bit addresses and byte-addressable memory, the following bits of the address are used to access the cache:

Tag	Index	Offset
31-10	9-5	4-0

a) What is the cache line size? (**2p**)
b) How many entries does the cache have? (**2p**)

Now assume the following code:

```
struct Student {
  unsigned Id;
  int Grades[15];
  char Name[128];
}
...
unsigned N = 100000
Student* Students = new Student[N];

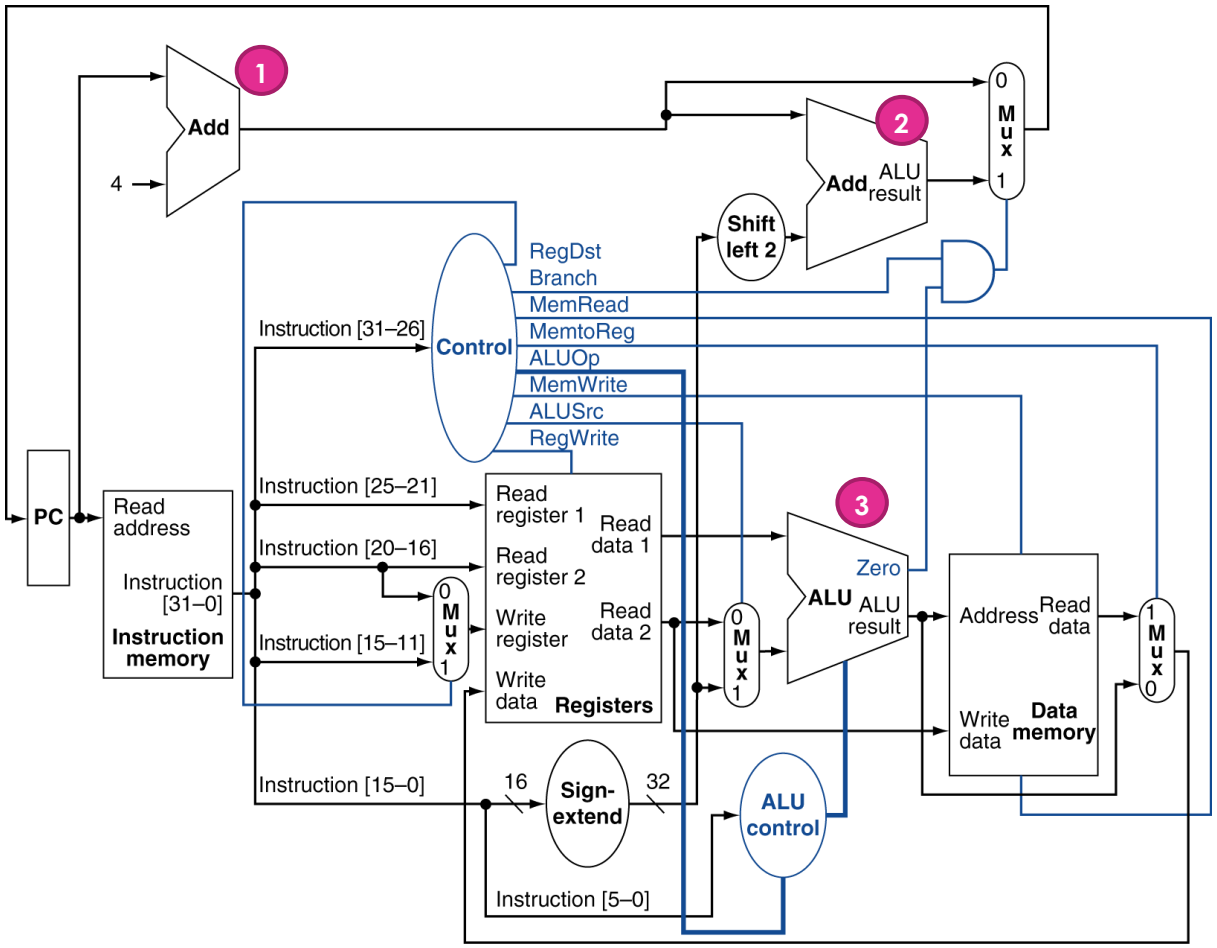

float getavg(unsigned id) {...}
```

In the rest of the program, the function getavg() gets a student Id as input and returns the average of his/her Grades. The array is not sorted by Ids. The cache is defined as above.

c) What is the number of cache misses running getavg () assuming that the matching entry is placed at the position ***i*** in the Students array, assuming $0<i<N$? (**3p**)
d) How can you modify the program to reduce the number of cache misses during the search? How many cache misses do you have now? (**5p**)

Task 6 (12p)

The following diagram shows a simple datapath for MIPS instruction set. The datapath contains 3 ALUs marked by number 1-3.



A. Explain the job of each of these ALUs in detail. Use examples. (**6p**)
B. The output of ALUs number 1 and 2 goes to a Mux. What is the purpose of this Mux? What is the other input to it? (**4p**)
C. ALU 3 has an output marked as "Zero". What does it represent and do? (**2p**)